Lecture 4 — OLS Asymptotics

Introductory Econometrics — Télécom Paris

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Reading: Wooldridge (5th ed.) chapter 5

Review: Finite-sample properties of OLS

The multiple linear regression model is

$$Y = \underbrace{X}_{n\times(K+1)} \quad \underbrace{\beta}_{(K+1)\times 1} \quad + \quad u.$$

Under MLR.1–MLR.4 (linearity, iid, full rank, $\mathbb{E}(u \mid X) = 0$): **unbiasedness of OLS**.

$$\hat{\beta} = (X'X)^{-1}X'Y, \qquad \mathbb{E}(\hat{\beta} \mid X) = \beta.$$

Under MLR.1–MLR.5 ($+\mathrm{Var}(u \mid X) = \sigma^2 I_n$): **Gauss–Markov theorem**. OLS is BLUE.

$$\mathrm{Var}(\hat{\beta} \mid X) = \sigma^2 (X'X)^{-1}.$$

The residual variance estimator is also unbiased:

$$\hat{\sigma}^2 = \frac{\hat{u}'\hat{u}}{n-(K+1)}, \qquad \mathbb{E}(\hat{\sigma}^2 \mid X) = \sigma^2.$$

These properties do not require normal errors or a limit as $n \to \infty$.

Review: Exact finite-sample distributions and inference

Under MLR.1–MLR.6 ($+u \mid X \sim N(0, \sigma^2 I_n)$): the conditional sampling distributions are exact for every finite $n > K + 1$:

$$\hat{\beta} \mid X \sim N\big(\beta, \sigma^2 (X'X)^{-1}\big)\,,$$

$$\hat{u} \mid X \sim N\big(0, \sigma^2 [I_n - X(X'X)^{-1}X']\big)\,, \qquad \frac{[n-(K+1)]\hat{\sigma}^2}{\sigma^2} \,\Big|\, X \sim \chi^2_{n-(K+1)},$$

$$\hat{\beta} \perp\!\!\!\perp \hat{\sigma}^2 \mid X.$$

These results give exact finite-sample t and F distributions under the null, and therefore exact hypothesis tests and confidence intervals.

Asymptotic results instead describe a limit as $n \to \infty$ and justify large-sample approximations.

From normal errors to large-sample inference

In the linear model, MLR.6 is equivalent to a normal conditional distribution of the outcome:

$$y \mid x_1, \ldots, x_K \sim N(\beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K, \sigma^2).$$

This cannot hold exactly for many outcomes. For example:

- Employment status takes only the values 0 and 1.
- The number of doctor visits takes nonnegative integer values.

Does losing this exact justification prevent us from assessing statistical significance?

Under the maintained assumptions and suitable moment conditions, the CLT implies that OLS estimators are approximately normally distributed in sufficiently large samples. This gives a basis for approximate inference without normal errors.

Semiparametric model

It is important to know the asymptotic (large-sample) properties of estimators and test statistics.

Maintained assumptions: iid sample, full rank, $\mathbb{E}(u \mid X) = 0$ and $\mathrm{Var}(u \mid X) = \sigma^2 I_n$.

We no longer specify the full conditional distribution of the errors (MLR.6).

Even then, OLS has satisfactory large-sample properties: under the null, t and F statistics have approximate t and F distributions, under the regularity conditions developed below.

The asymptotic experiment

Observe an iid sample of n observations from the same population:

$$(y_1, x_1), \ldots, (y_n, x_n).$$

The population parameters and the number of regressors remain fixed.

For each n, $\hat{\beta}_n$ has a probability distribution over the values it could take in different random samples of size n.

1. **Unbiasedness:** is the mean of the sampling distribution equal to β_j at a given n?
2. **Consistency:** does the distribution of $\hat{\beta}_{n,j}$ concentrate near β_j as n grows?
3. **Asymptotic normality:** what is the limiting distribution of the scaled estimation error?
4. **Inference and efficiency:** how do we quantify uncertainty and compare estimators?

Consistency

Consistency: Modes of convergence

Let $(U_n)_{n\in\mathbb{N}}$ and U be random vectors of the same fixed dimension.

Definition. U_n **converges in quadratic mean** to U, denoted $U_n \xrightarrow{L^2} U$, if

$$\lim_{n\to\infty} \mathbb{E}[\|U_n - U\|^2] = 0.$$

Definition. U_n **converges in probability** to U, denoted $U_n \xrightarrow{p} U$, if for all $\varepsilon > 0$,

$$\lim_{n\to\infty} \Pr(\|U_n - U\| > \varepsilon) = 0.$$

Equivalently, $U_n - U = o_p(1)$. For a constant c, write $\operatorname{plim} U_n = c$.

Consistency: From quadratic mean to probability

Proposition. Convergence in quadratic mean implies convergence in probability:

$$U_n \xrightarrow{L^2} U \quad \Longrightarrow \quad U_n \xrightarrow{p} U.$$

Proof. Markov's inequality applied to $\|U_n - U\|^2$ gives, for every $\varepsilon > 0$,

$$\Pr(\|U_n - U\| > \varepsilon) \leq \frac{\mathbb{E}[\|U_n - U\|^2]}{\varepsilon^2} \xrightarrow[n\to\infty]{} 0.$$

For scalar U_n centered at its mean, this is Chebyshev's variance bound.

A useful sufficient condition. For a scalar U_n and a constant c,

$$\mathbb{E}[(U_n - c)^2] = \mathrm{Var}(U_n) + [\mathbb{E}(U_n) - c]^2.$$

Vanishing variance and bias therefore imply mean-square convergence to c, and hence convergence in probability to c.

Consistency: Weak law of large numbers

Theorem (Weak Law of Large Numbers). Let $(U_i)_{i\in\mathbb{N}}$ be iid random variables with finite absolute first moment, then

$$\frac{1}{n}\sum_{i=1}^{n} U_i \xrightarrow[n\to\infty]{p} \mathbb{E}(U_i).$$

For scalars, the moment condition is $\mathbb{E}|U_i| < \infty$. For vectors and matrices, apply the scalar WLLN to each entry; with fixed dimensions, these limits hold jointly.

Consistency: Bernoulli illustration

Example. A Bernoulli distribution with parameter $p = 0.7$:

$$\Pr(U_i = 1) = 0.7, \qquad \Pr(U_i = 0) = 0.3.$$

Its mean and variance are

$$\mathbb{E}(U_i) = p, \qquad \mathrm{Var}(U_i) = p(1-p).$$

The sample mean $\bar{U}_n = \dfrac{1}{n}\sum_{i=1}^{n} U_i$ takes its values in

$$\left\{0, \frac{1}{n}, \ldots, \frac{n-1}{n}, 1\right\}.$$

Consistency: Bernoulli illustration

Example (continued).

$$\begin{aligned}\Pr\left(\bar{U}_n = \frac{k}{n}\right) &= \Pr(k \text{ variables equal 1 and } n-k \text{ variables equal 0}) \\ &= \binom{n}{k} p^k (1-p)^{n-k} \\ &= \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}.\end{aligned}$$

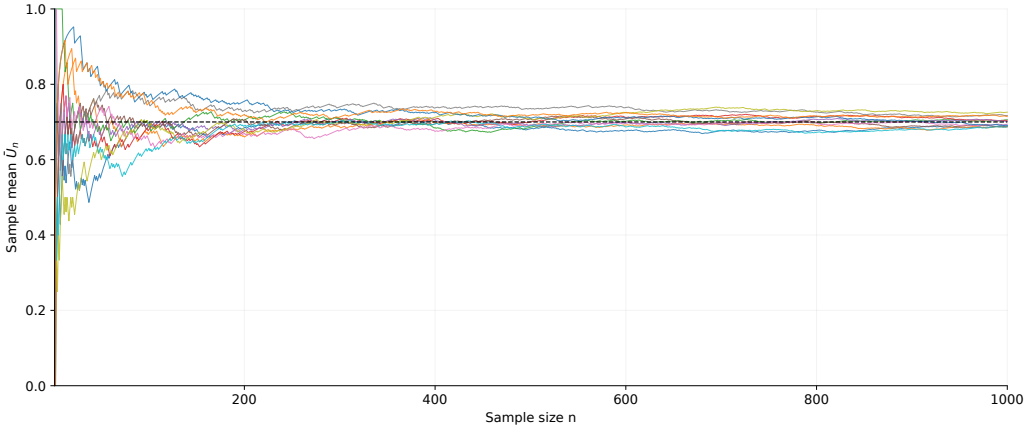
Thus $n\bar{U}_n$ has a binomial distribution.

The weak law of large numbers gives

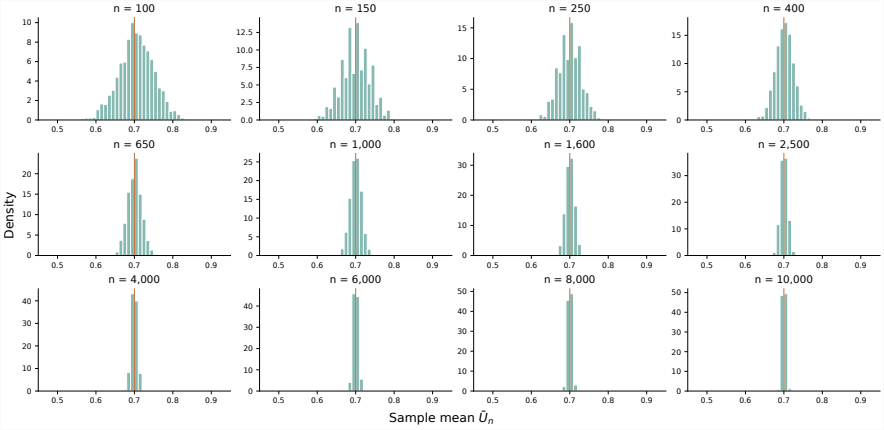
$$\bar{U}_n \xrightarrow[n\to+\infty]{p} p.$$

For a very large number of Bernoulli draws, the distribution of their sample proportion concentrates at a constant.

Paths of simulated sample means as the sample size increases



Sample-mean distributions for n from 100 to 10,000



Bernoulli(0.7) draws; 2,000 repetitions for each sample size.

Consistency: Continuous mapping theorem

Theorem (Continuous Mapping Theorem). If $W_n \xrightarrow{p} w$, where w is constant, and g is continuous at w, then

$$g(W_n) \xrightarrow{p} g(w).$$

The statement applies to fixed-dimensional vectors and matrices as well as scalars.

Probability-limit rules. If $A_n \xrightarrow{p} A$ and $B_n \xrightarrow{p} B$ with constant limits, then $A_n + B_n \xrightarrow{p} A + B$ and $A_n B_n \xrightarrow{p} AB$ for compatible dimensions. For scalars, $A_n / B_n \xrightarrow{p} A/B$ if $B \neq 0$. Independence is not required.

Application I: If $W_n \xrightarrow{p} \mu$ in $\mathbb{R}^p$, then $a' W_n \xrightarrow{p} a'\mu$ for every fixed $a \in \mathbb{R}^p$.

Application II: If $Q_n \xrightarrow{p} Q$ and Q is nonsingular, then

$$Q_n^{-1} \xrightarrow{p} Q^{-1},$$

because inversion is continuous at every nonsingular matrix.

Consistency

Let $Y_i = X_i\beta + u_i$, with $X_i = (1, x_{i1}, \ldots, x_{iK})$ a row vector and $K+1$ fixed.

Under assumptions:

1. (X_i, u_i) are iid, $\mathbb{E}\|X_i\|^2 < \infty$ and $\mathbb{E}(u_i^2) < \infty$.
2. The population matrix $Q = \mathbb{E}(X_i' X_i)$ is positive definite: $a'Qa = \mathbb{E}[(X_i a)^2] > 0$ for every nonzero $a \in \mathbb{R}^{K+1}$.
3. $\mathbb{E}(u_i \mid X_i) = 0$, hence $\mathbb{E}(X_i' u_i) = 0$ by iterated expectations.

Theorem (Consistency of OLS). The OLS estimator is consistent:

$$\hat{\beta}_n \xrightarrow{p} \beta.$$

Consistency: Proof—the exact estimation error

Step 1: $\hat{\beta}_n$ can be rewritten as:

$$\hat{\beta}_n = (X'X)^{-1}X'Y = \beta + (X'X)^{-1}X'u.$$

Write both sums as sample averages:

$$Q_n = \frac{X'X}{n} = \frac{1}{n}\sum_i X_i'X_i, \qquad g_n = \frac{X'u}{n} = \frac{1}{n}\sum_i X_i'u_i.$$

Then

$$\boxed{\hat{\beta}_n - \beta = Q_n^{-1}g_n.}$$

We must show that the inverse regressor moment matrix stabilizes and the regressor–error moment tends to zero.

Consistency: Proof—sample moments

Step 2: apply the WLLN to each entry. Write $X_i = (x_{i0}, \dots, x_{iK})$, where $x_{i0} = 1$. Then

$$(X'X)_{ab} = \sum_{i=1}^{n} x_{ia}x_{ib}, \qquad (X'u)_a = \sum_{i=1}^{n} x_{ia}u_i, \quad a, b = 0, \dots, K.$$

Finite second moments make these products integrable by Cauchy–Schwarz. The scalar WLLN gives

$$\frac{1}{n}\sum_i x_{ia}x_{ib} \xrightarrow{p} \mathbb{E}(x_{ia}x_{ib}), \qquad \frac{1}{n}\sum_i x_{ia}u_i \xrightarrow{p} \mathbb{E}(x_{ia}u_i) = 0.$$

Since $K + 1$ is fixed, combining the entrywise limits gives

$$Q_n = \frac{1}{n}\sum_i X_i'X_i \xrightarrow{p} \mathbb{E}(X_i'X_i) = Q, \qquad g_n = \frac{1}{n}\sum_i X_i'u_i \xrightarrow{p} \mathbb{E}(X_i'u_i) = 0.$$

The zero limit uses **orthogonality**, implied by zero conditional mean.

Consistency: Proof—combine the limits

Step 3: continuous mapping. Population full rank makes Q nonsingular, so

$$Q_n \xrightarrow{p} Q \quad \Longrightarrow \quad Q_n^{-1} \xrightarrow{p} Q^{-1}.$$

Step 4: the product rule for probability limits.

$$\hat{\beta}_n - \beta = Q_n^{-1} g_n \xrightarrow{p} Q^{-1} 0 = 0.$$

Conclusion: $\hat{\beta}_n \xrightarrow{p} \beta$. Each OLS coefficient consistently estimates its population value.

This establishes concentration, not the shape of the sampling distribution. The next goal is the asymptotic distribution of $\sqrt{n}(\hat{\beta}_n - \beta)$.

Proof: Consistency of the sample variance

For iid Z_i with $\mathbb{E}(Z_i) = \mu$ and $0 < \mathrm{Var}(Z_i) = \tau^2 < \infty$, we want to show that $s_n^2 \xrightarrow{p} \tau^2$.

1. Rewrite the sample variance in terms of sample moments:

$$s_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Z_i - \bar{Z}_n)^2 = \frac{n}{n-1}\left(\frac{1}{n}\sum_{i=1}^{n} Z_i^2 - \bar{Z}_n^2\right).$$

2. Apply the LLN to both moments: finite variance gives $\mathbb{E}(Z_i^2) < \infty$.

$$\bar{Z}_n \xrightarrow{p} \mu, \qquad \frac{1}{n}\sum_{i=1}^{n} Z_i^2 \xrightarrow{p} \mathbb{E}(Z_i^2).$$

3. Combine the limits: $\bar{Z}_n^2 \xrightarrow{p} \mu^2$ by continuous mapping, and $n/(n-1) \to 1$. Hence

$$s_n^2 \xrightarrow{p} \mathbb{E}(Z_i^2) - \mu^2 = \tau^2.$$

The sample variance is consistent; we will use this to justify replacing an unknown variance in large-sample inference. No fourth moment is needed.

Consistency: Finite-sample and large-sample properties

Unbiasedness concerns an expectation at a given sample size:

$$\mathbb{E}(\hat{\theta}_n) = \theta.$$

Consistency concerns concentration as the sample grows:

$$\hat{\theta}_n \xrightarrow{p} \theta.$$

An estimator can be centered correctly at every n without concentrating. A biased estimator can nevertheless concentrate on the correct parameter.

Asymptotic Distribution

Asymptotic Distribution: Convergence in distribution

Convergence in distribution. A sequence of random variables $(U_n)_{n\in\mathbb{N}}$ with cumulative distribution functions $(F_{U_n}(\cdot))_{n\in\mathbb{N}}$ converges in distribution if there exists a random variable U such that

$$\lim_{n\to\infty} F_{U_n}(a) = F_U(a)$$

at every continuity point a of F_U.

Notation: $U_n \xrightarrow{d} U$, $U_n \xrightarrow{\mathcal{L}} U$, or $F_{U_n} \longrightarrow F_U$.

Propositions:

$$U_n \xrightarrow{p} U \quad \Longrightarrow \quad U_n \xrightarrow{d} U,$$

$$U_n \xrightarrow{d} c \text{ with constant } c \quad \Longrightarrow \quad U_n \xrightarrow{p} c.$$

Asymptotic Distribution: Central limit theorem

Theorem (Central Limit Theorem). Let $(U_i)_{i\in\mathbb{N}}$ be a sequence of iid real random variables with $\mathbb{E}(U_i^2) < \infty$. For $S_n = \frac{1}{n}\sum_{i=1}^{n} U_i$, the scaled deviation $\sqrt{n}[S_n - \mathbb{E}(U_i)]$ converges in distribution to a normal random variable with mean zero and variance $\mathrm{Var}(U_i)$:

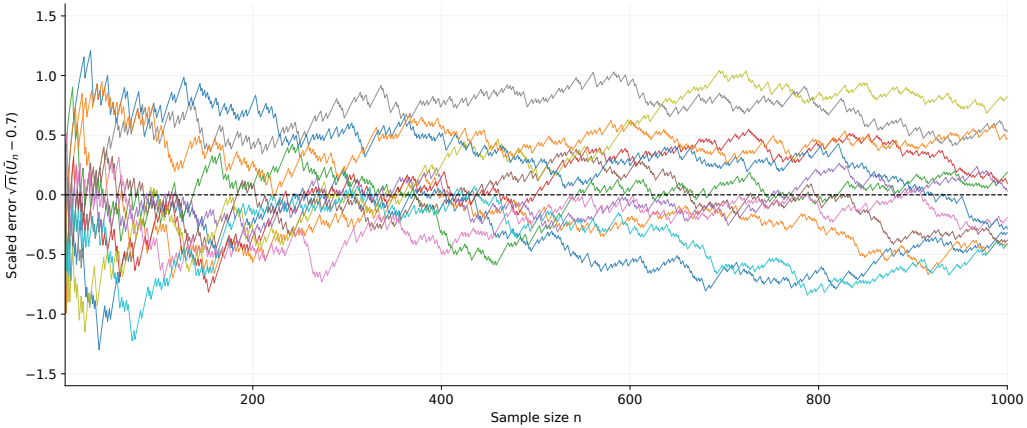
$$\sqrt{n}[S_n - \mathbb{E}(U_i)] \xrightarrow{\mathcal{L}} N(0, \mathrm{Var}(U_i)).$$

Equivalently, when $\mathrm{Var}(U_i) > 0$, $\quad \Pr\left(\frac{\sqrt{n}[S_n - \mathbb{E}(U_i)]}{\sqrt{\mathrm{Var}(U_i)}} \leq x\right) \xrightarrow[n\to+\infty]{} \Phi(x), \quad \forall x \in \mathbb{R},$

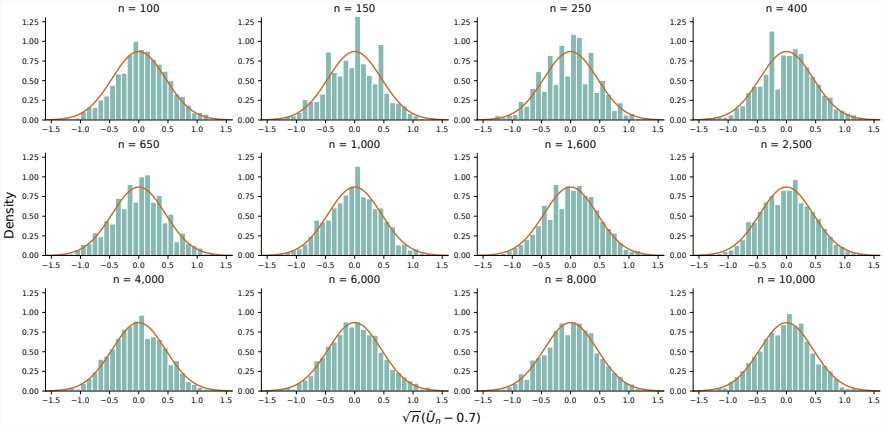
where Φ is the cumulative distribution function of the standard normal distribution.

The normal limit describes the scaled sampling error of the mean.

Paths of deviations from the true value scaled by $\sqrt{n}$



Distributions of deviations from the true value scaled by $\sqrt{n}$



Bernoulli(0.7); 2,000 repetitions. Orange curve: $N(0, 0.7 \times 0.3)$ density.

Asymptotic Distribution: Multivariate CLT

Theorem (Multivariate Central Limit Theorem). Let $(U_i)_{i\in\mathbb{N}}$ be iid random vectors of dimension p, with $\mathbb{E}(U_{i\ell}^2) < \infty$ for $\ell = 1, \ldots, p$.

For $S_n = \frac{1}{n}\sum_{i=1}^{n} U_i$, the scaled deviation $\sqrt{n}[S_n - \mathbb{E}(U_i)]$ converges in distribution to a multivariate normal random variable with mean zero and variance–covariance matrix

$$\Sigma = \mathrm{Var}(U_i) = \mathbb{E}[(U_i - \mathbb{E}U_i)(U_i - \mathbb{E}U_i)'].$$

$$\sqrt{n}[S_n - \mathbb{E}(U_i)] \xrightarrow{\mathcal{L}} N_p(0, \Sigma).$$

Asymptotic Distribution: Continuous mapping

Theorem (Continuous Mapping Theorem). If $W_n \xrightarrow{d} W$ and g is continuous at every point in a set to which W belongs with probability one, then

$$g(W_n) \xrightarrow{d} g(W).$$

Application: for a fixed matrix B of compatible dimensions,

$$W_n \xrightarrow{d} W \quad \Longrightarrow \quad BW_n \xrightarrow{d} BW.$$

If $W \sim N_p(0, \Sigma)$, then $BW \sim N_q(0, B\Sigma B')$ for a $q \times p$ matrix B.

Asymptotic Distribution: Slutsky's theorem

Theorem (Slutsky's Theorem). If $Z_n \xrightarrow{d} Z$ and $A_n \xrightarrow{p} A$, where A is constant, then, for compatible fixed dimensions,

$$A_n Z_n \xrightarrow{d} AZ,$$

$$Z_n + A_n \xrightarrow{d} Z + A.$$

For a scalar $A \neq 0$, $Z_n / A_n \xrightarrow{d} Z/A$.

Independence between A_n and Z_n is not required. The constant probability limit is essential.

This lets us combine a matrix converging in probability with an estimation error converging in distribution.

Asymptotic Distribution: OLS theorem

Retain the consistency assumptions: iid sampling, fixed $K+1$, finite second moments, positive definite $Q = \mathbb{E}(X_i' X_i)$, and $\mathbb{E}(u_i \mid X_i) = 0$.

Assume additionally $\mathrm{Var}(u_i \mid X_i) = \sigma^2$, with $0 < \sigma^2 < \infty$.

Theorem (Asymptotic Normality of OLS). The scaled estimation error satisfies

$$\sqrt{n}(\hat{\beta}_n - \beta) \xrightarrow{d} N_{K+1}(0, \sigma^2 Q^{-1}).$$

The limit describes the scaled error. It gives the large-sample approximation

$$\hat{\beta}_n \overset{a}{\sim} N_{K+1}\left(\beta, \frac{\sigma^2}{n} Q^{-1}\right).$$

Normality of the errors is not required.

Asymptotic Distribution: Proof—scaled error

Step 1: rewrite the scaled error. From the consistency proof,

$$\hat{\beta}_n - \beta = Q_n^{-1} g_n, \qquad Q_n = \frac{X'X}{n}, \quad g_n = \frac{X'u}{n}.$$

Multiplying by $\sqrt{n}$ gives

$$\sqrt{n}(\hat{\beta}_n - \beta) = \underbrace{Q_n^{-1}}_{\xrightarrow{p} Q^{-1}} \underbrace{\frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i' u_i}_{Z_n}.$$

The first limit follows from the **WLLN and continuous mapping**, already established in the consistency proof.

We now use the **multivariate CLT** to obtain the distributional limit of Z_n.

Asymptotic Distribution: Proof—mean and covariance

Step 2: find the moments needed for the CLT. Let $V_i = X_i' u_i \in \mathbb{R}^{K+1}$. The V_i are iid, and zero conditional mean gives

$$\mathbb{E}(V_i) = \mathbb{E}[X_i' \mathbb{E}(u_i \mid X_i)] = 0.$$

Zero conditional mean and homoskedasticity give $\mathbb{E}(u_i^2 \mid X_i) = \sigma^2$. Therefore

$$\mathrm{Var}(V_i) = \mathbb{E}(V_i V_i') = \mathbb{E}(u_i^2 X_i' X_i) = \mathbb{E}[X_i' X_i \mathbb{E}(u_i^2 \mid X_i)] = \sigma^2 Q.$$

Every component has finite variance: $\mathbb{E}(x_{ia}^2 u_i^2) = \sigma^2 \mathbb{E}(x_{ia}^2) < \infty$. Thus the finite-second-moment condition of the multivariate CLT is satisfied.

Asymptotic Distribution: Proof—apply the CLT

Step 3: apply the multivariate CLT to $V_i = X_i' u_i$. Since $\mathbb{E}(V_i) = 0$ and $\mathrm{Var}(V_i) = \sigma^2 Q$,

$$Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} V_i = \sqrt{n} \left(\frac{1}{n} \sum_{i=1}^{n} V_i - 0 \right) \xrightarrow{d} Z, \qquad Z \sim N_{K+1}(0, \sigma^2 Q).$$

The CLT applies to the **regressor–error products** $X_i' u_i$, not directly to the coefficient estimator.

It is a joint vector limit: the components need not be independent, and their covariances are recorded in $\sigma^2 Q$.

Asymptotic Distribution: Proof—combine the limits

Step 4: Slutsky's theorem. Combine $Q_n^{-1} \xrightarrow{p} Q^{-1}$ with $Z_n \xrightarrow{d} Z$:

$$\sqrt{n}(\hat{\beta}_n - \beta) = Q_n^{-1} Z_n \xrightarrow{d} Q^{-1} Z.$$

Step 5: transform the normal covariance. Since Q is symmetric,

$$\mathrm{Var}(Q^{-1}Z) = Q^{-1}(\sigma^2 Q)(Q^{-1})' = \sigma^2 Q^{-1}.$$

Therefore

$$\boxed{\sqrt{n}(\hat{\beta}_n - \beta) \xrightarrow{d} N_{K+1}(0, \sigma^2 Q^{-1}).}$$

The LLN stabilized the regressor moment matrix; the CLT supplied the random limit; Slutsky combined them.

Estimating the asymptotic variance

We already have $Q_n = X'X/n \xrightarrow{p} Q$ by the LLN, and $Q_n^{-1} \xrightarrow{p} Q^{-1}$ by the CMT because Q is nonsingular.

It remains to estimate σ^2. The residual sum of squares satisfies

$$\sum_i \hat{u}_i^2 = \hat{u}'\hat{u} = u'[I_n - X(X'X)^{-1}X']u.$$

Thus

$$\hat{\sigma}_n^2 = \frac{n}{n-(K+1)} \left[\frac{u'u}{n} - \frac{u'X}{n} \left(\frac{X'X}{n} \right)^{-1} \frac{X'u}{n} \right].$$

Under iid, ZCM, homoskedasticity and finite-moment assumptions, the LLN gives

$$\frac{u'u}{n} \xrightarrow{p} \mathbb{E}(u_i^2) = \sigma^2, \qquad \frac{X'u}{n} \xrightarrow{p} 0.$$

With fixed K, $n/(n-K-1) \to 1$, so $\hat{\sigma}_n^2 \xrightarrow{p} \sigma^2$.

Consequences: Estimated covariance and standard errors

By the CMT, $\hat{\sigma}_n^2 \xrightarrow{p} \sigma^2 > 0$ implies $\hat{\sigma}_n \xrightarrow{p} \sigma$. Slutsky combines this with the preceding normal limit:

$$\frac{\sqrt{n}(\hat{\beta}_n - \beta)}{\hat{\sigma}_n} \xrightarrow{d} N_{K+1}(0, Q^{-1}).$$

Implications. Let $Q_n = X'X/n$.

- Estimated asymptotic covariance of $\sqrt{n}(\hat{\beta}_n - \beta)$: $\widehat{V}_\infty = \hat{\sigma}_n^2 Q_n^{-1} \xrightarrow{p} \sigma^2 Q^{-1}$.
- Estimated covariance for $\hat{\beta}_n$: $\widehat{V} = \widehat{V}_\infty / n = \hat{\sigma}_n^2 (X'X)^{-1}$.
- Standard error of coefficient j: $\text{se}(\hat{\beta}_{n,j}) = \sqrt{\widehat{V}_{jj}}$.

For $v_j = (\sigma^2 Q^{-1})_{jj} > 0$, $n\widehat{V}_{jj} \xrightarrow{p} v_j$. Slutsky therefore gives

$$\frac{\hat{\beta}_{n,j} - \beta_j}{\text{se}(\hat{\beta}_{n,j})} = \frac{\sqrt{n}(\hat{\beta}_{n,j} - \beta_j)}{\sqrt{n\widehat{V}_{jj}}} \xrightarrow{d} N(0,1).$$

This is the result used for individual coefficient tests and confidence intervals.

Inference

Inference: Use the familiar t and F procedures

Under the maintained iid, rank, zero conditional mean, homoskedasticity and moment conditions, the preceding results justify the familiar inference procedures in large samples.

Individual hypotheses and confidence intervals. t testing and the construction of confidence intervals are carried out exactly as under the classical linear model assumptions: use the same statistics, standard errors and t_{n-K-1} critical values.

Without normal errors, the t reference distribution is an approximation, justified asymptotically by the standard normal limit established above.

Multiple hypotheses. The usual F statistic has an approximate $F_{q,n-K-1}$ distribution under the null in large samples, for a fixed number q of restrictions. For exclusion restrictions or other linear restrictions, use the same procedure as in Lecture 3.

What changes is the guarantee: inference is approximate in large samples, rather than exact at every finite sample size under conditional normality.

Inference: When can the approximation be poor?

If the sample size is not very large and the errors are not normally distributed, the t distribution can be a poor approximation to the distribution of the t statistic.

The error distribution matters. Strong skewness or heavy tails can require more observations before the central limit theorem delivers a useful approximation, provided the required moments exist.

The model size matters. With more explanatory variables, fewer residual degrees of freedom remain. A larger sample is usually needed for the t approximation to work well.

There is no universal sample-size threshold: approximation quality also depends on the regressor distribution and whether a few observations dominate the estimates.

Other tests: The Lagrange multiplier statistic

Asymptotic analysis also provides other ways to test multiple exclusion restrictions. Consider our standard model $Y_i = X_{1,i}\beta_1 + X_{2,i}\beta_2 + u_i$, where $X_{2,i}$ contains the last q regressors. Test

$$H_0 : \beta_2 = 0 \qquad \text{against} \qquad H_1 : \exists\, k \in \{1, \ldots, q\} : \beta_{2,k} \neq 0.$$

Key idea: under H_0, restricted residuals should be approximately uncorrelated with each omitted variable in the sample.

There are q exclusion restrictions. Under the maintained homoskedastic model:

1. Regress Y on X_1 and obtain $\tilde{u} = Y - X_1\tilde{\beta}_1$.
2. Regress $\tilde{u}$ on all variables in X_1 and X_2; record the auxiliary R^2. Under H_0, it should be close to zero: $\tilde{u}$ is approximately uncorrelated with all regressors.
3. Under H_0, $LM = nR^2 \xrightarrow{d} \chi^2_q$.
4. Reject H_0 if $LM > c_{q,1-\alpha}$ (the upper-tail α critical value), or if the p-value $\Pr(\chi^2_q \geq LM_{\text{obs}}) < \alpha$.

What the LM auxiliary regression measures

Partition $X = [X_1\ X_2]$, where X_1 contains an intercept and X_2 contains the q excluded regressors.

Restricted residuals $\tilde{u} = Y - X_1\tilde{\beta}_1$ are orthogonal to X_1 by construction. Under $H_0 : \beta_2 = 0$, the excluded variables should not systematically explain these residuals.

In the auxiliary regression of $\tilde{u}$ on $[X_1\ X_2]$,

$$R^2_{\text{aux}} = 1 - \frac{\text{SSR}_{UR}}{\text{SSR}_R}, \qquad LM = nR^2_{\text{aux}}.$$

Use the auxiliary R^2, not the R^2 of the original outcome regression.

Conclusion

- Under the usual linear-model assumptions (iid sampling, zero conditional mean, population rank and suitable moments), OLS is consistent as the sample size grows.

- Without normally distributed errors, and retaining homoskedasticity:
  - Student's statistics are asymptotically standard normal.
  - Appropriately scaled Fisher statistics satisfy $qF_n \xrightarrow{d} \chi^2_q$, where q is the number of restrictions.
  - Asymptotic confidence intervals with coverage $1 - \alpha$ can be constructed using normal quantiles.

Summary and the next modelling question

The LLN and continuous mapping locate the OLS probability limit. The CLT and Slutsky provide the scaled distribution and feasible inference.

Large samples can relax normality. They do not eliminate omitted-variable inconsistency or establish a causal interpretation.

Next we study how to specify useful conditional relationships: curvature, categories and interactions.

Reading: Wooldridge (5th ed.), §5.1–5.3; Appendix C.3 for asymptotic concepts and Appendix E.4 for the matrix argument.